



Course Outline

Course Objective and Outcome Form
Department of Electrical and Computer Engineering
School of Engineering and Physical Sciences
North South University, Bashundhara, Dhaka-1229, Bangladesh

1. Course Number and Title

CSE465 – Pattern Recognition
and Neural Networks

2. Semester: *Optional*

3. Number of Credits

3 Credits (Theory)

4. Type

Core, Engineering, Lecture

5. Prerequisites

None

6. Contact Hours

Lecture – 3 hours per week (Total 36 hours)

7. Course Summary

Introduction to pattern matching algorithms. In this course, we will focus on neural network based methods of pattern recognition only. We will start with deep feedforward neural networks. Then we will explore how we can customize neural networks for different kind of problem solving: for image/video based problems, for time series, for natural language processing, etc.

8. Course Objectives

- The course will cover topics such as supervised learning, unsupervised learning mainly in the context of neural networks.
- We will cover techniques, mainly popularised in the last few years, for neural network training
- We aim to give students an appreciation for the machine learning process involving this class of models.

9. Course Outcomes (COs)

CO	Course Outcome Description	Weightage (%)
CO1:	Gain mathematical and judgmental insights into the deep learning algorithms	20
CO2:	Be able to pick the most appropriate algorithm for a problem	30
CO3:	Be able to use a tensor-based framework to implement DL projects	50

10. Mapping of CO-PO

CO	Mapped POs	KP	Bloom's Level	Delivery Methods	Assessment Tools
CO1	PO1	K3	C3 – Apply	Lectures, Notes	Quiz / Midterm
CO2	PO1	K3	C4 – Analyze	Lectures, Notes	Quiz/Midterm / Final Exam
CO3	PO1	K3	C4 – Analyze	Lectures, Notes	Project

11. Resources

- Textbooks
 - <https://www.deeplearningbook.org/>
 - <http://neuralnetworksanddeeplearning.com/>

(You will be expected to read a lot of academic papers)

12. Weekly Course Schedule (Theory)

(12 Weeks = 36 Contact Hours)

Week#: Topics

Week 1: Introduction to Deep Learning and Mathematical Foundations

- Topics:
 - Overview of deep learning, history, and applications
 - Differences between traditional ML and deep learning
 - Linear algebra refresher: vectors, matrices, dot products
 - Introduction to Python, NumPy, and PyTorch/TensorFlow

• Learning Outcomes:

- Understand the significance of deep learning in AI.
- Review basic mathematical concepts used in deep learning.
- Set up an environment for deep learning development.
- Assignment: Implement a simple linear regression using NumPy.
- Suggested Readings:
 - *Deep Learning* by Ian Goodfellow, Chapter 1
 - *Neural Networks and Deep Learning* by Michael Nielsen, Chapter 1

Week 2: Neural Networks Basics

- **Topics:**
 - Introduction to artificial neurons and architecture
 - Building blocks of neural networks: layers, activation functions
 - Loss functions and backpropagation
 - **Learning Outcomes:**
 - Understand how a neural network is structured and how it learns.
 - Understand the role of different activation functions and loss functions.
 - **Assignment:** Build a simple neural network from scratch to classify a toy dataset (e.g., Iris dataset).
 - **Suggested Readings:**
 - *Deep Learning* by Ian Goodfellow, Chapter 2
 - *Neural Networks and Deep Learning* by Michael Nielsen, Chapter 2
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Week 3: Training Neural Networks

- **Topics:**
 - Optimization algorithms: Gradient Descent, Stochastic Gradient Descent, Adam
 - Weight initialization techniques
 - Gradient vanishing and exploding problems
 - **Learning Outcomes:**
 - Understand how neural networks are trained using optimization techniques.
 - Learn how to handle common training challenges.
 - **Assignment:** Train a simple neural network with different optimization techniques (SGD vs Adam).
 - **Suggested Readings:**
 - *Deep Learning* by Ian Goodfellow, Chapter 6
 - *Neural Networks and Deep Learning* by Michael Nielsen, Chapter 3
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Week 4: Regularization and Generalization

- **Topics:**
 - Overfitting and underfitting
 - Regularization methods: L1/L2 regularization, dropout
 - Cross-validation and early stopping
- **Learning Outcomes:**
 - Learn strategies to improve model generalization.
 - Understand and implement regularization techniques.
- **Assignment:** Implement dropout and L2 regularization in a neural network.
- **Suggested Readings:**
 - *Deep Learning* by Ian Goodfellow, Chapter 7

- *Neural Networks and Deep Learning* by Michael Nielsen, Chapter 4
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Week 5: Introduction to Convolutional Neural Networks (CNNs)

- **Topics:**
 - Convolution and pooling layers
 - Architecture of CNNs: LeNet, AlexNet
 - Applications in image classification
 - **Learning Outcomes:**
 - Understand the building blocks and architecture of CNNs.
 - Learn how CNNs are used in computer vision.
 - **Assignment:** Implement a simple CNN to classify MNIST dataset.
 - **Suggested Readings:**
 - *Deep Learning* by Ian Goodfellow, Chapter 9
 - *Neural Networks and Deep Learning* by Michael Nielsen, Chapter 6
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Week 6: Advanced CNN Architectures

- **Topics:**
 - Deep CNNs: VGGNet, ResNet
 - Batch normalization
 - Data augmentation and transfer learning
 - **Learning Outcomes:**
 - Explore advanced CNN architectures and understand their improvements.
 - Implement transfer learning with pre-trained models.
 - **Assignment:** Fine-tune a pre-trained ResNet model for a new classification task.
 - **Suggested Readings:**
 - *Deep Learning* by Ian Goodfellow, Chapter 9
 - *Convolutional Neural Networks* (CS231n) lectures by Stanford
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Week 7: Introduction to Natural Language Processing (NLP)

- **Topics:**
 - Basics of NLP: Tokenization, stemming, and lemmatization
 - Bag-of-Words (BoW) and TF-IDF
 - Word embeddings: Word2Vec, GloVe
- **Learning Outcomes:**
 - Understand fundamental NLP concepts and techniques.
 - Learn how text data is processed for neural networks.

- **Assignment:** Implement word embeddings using Word2Vec and visualize word vectors.
 - **Suggested Readings:**
 - *Deep Learning* by Ian Goodfellow, Chapter 10
 - *Speech and Language Processing* by Jurafsky and Martin, Chapter 2
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Week 8: Recurrent Neural Networks (RNNs) for Sequence Modeling

- **Topics:**
 - Introduction to RNNs, LSTM, and GRU
 - Applications: Text generation, language modeling
 - Vanishing gradient problem in RNNs
 - **Learning Outcomes:**
 - Understand how RNNs process sequences and temporal data.
 - Learn about LSTMs and GRUs to overcome RNN limitations.
 - **Assignment:** Build a simple text generator using LSTM.
 - **Suggested Readings:**
 - *Deep Learning* by Ian Goodfellow, Chapter 10
 - *Neural Networks and Deep Learning* by Michael Nielsen, Chapter 6
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Week 9: Attention Mechanism and Transformers

- **Topics:**
 - Introduction to attention mechanisms
 - The Transformer architecture
 - Applications in NLP: Machine translation, summarization
 - **Learning Outcomes:**
 - Understand the core principles of attention and transformers.
 - Learn how transformers revolutionized NLP.
 - **Assignment:** Implement a simple transformer-based model for text classification.
 - **Suggested Readings:**
 - *Attention is All You Need* (Vaswani et al.)
 - *Deep Learning* by Ian Goodfellow, Chapter 11
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Week 10: Generative Models in Deep Learning

- **Topics:**
 - Introduction to Generative Adversarial Networks (GANs)
 - Applications of GANs in image generation
 - Autoencoders and Variational Autoencoders (VAEs)
- **Learning Outcomes:**

- Understand the basic principles of GANs and VAEs.
 - Learn how to generate new data using deep learning.
- **Assignment:** Implement a simple GAN to generate MNIST digits.
- **Suggested Readings:**
 - *Deep Learning* by Ian Goodfellow, Chapter 20
 - *Neural Networks and Deep Learning* by Michael Nielsen, Chapter 7

Week 11: Model Evaluation and Metrics

- **Topics:**
 - Metrics for classification: Accuracy, precision, recall, F1-score
 - Confusion matrix and ROC curves
 - Evaluating model performance on real-world datasets
- **Learning Outcomes:**
 - Learn how to evaluate the effectiveness of deep learning models.
 - Understand various performance metrics and their importance.
- **Assignment:** Evaluate the performance of a CNN on a real-world dataset (e.g., CIFAR-10).
- **Suggested Readings:**
 - *Deep Learning* by Ian Goodfellow, Chapter 5
 - *Pattern Recognition and Machine Learning* by Christopher Bishop, Chapter 1

Week 12: Final Project and Future Directions

- **Topics:**
 - Recap of the course topics
 - Student presentations of final projects
 - Ethical implications of AI and the future of deep learning
- **Learning Outcomes:**
 - Consolidate learning from the entire course.
 - Discuss the future and ethical considerations of AI models.
- **Assignment:** Final project submission (integrate multiple deep learning techniques).
- **Suggested Readings:**
 - *Deep Learning* by Ian Goodfellow, Chapter 21 (Future Directions)
 - *Ethics of AI and Big Data* by K. Crawford

13. Weightage Distribution among Assessment Tools

Assessment Tools	Weightage (%)
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Paper Presentation + Quiz	10+10
Project	40
Midterm	20
Final Exam	20

14. Grading Policy

As per NSU grading policy available at:

www.northsouth.edu/academic/grading-policy.html